

Part A
(25 marks)

1. Find the following limits

a. $\lim_{x \rightarrow -5} \frac{25-x^2}{x+5}$ [2 Marks]

b. $\lim_{x \rightarrow 4} \frac{x-4}{\sqrt{x}-2}$ [4 Marks]

c. $\lim_{x \rightarrow \infty} \sqrt{\frac{3x+1}{6x-3}}$ [4 Marks]

2. Find $\frac{dy}{dx}$ of the following cases (Leave your final answer in the simplest form)

i. $y = e^{3x^2-2}$ [2 Marks]

ii. $y = \ln(\cos 2x - \sin 2x)$ [3 Marks]

iii. $ax^2 + 2hxy + by^2 = 1$, a,b are constants [5 Marks]

3. Given a function $= x^{\frac{5}{3}} + 10x^{\frac{2}{3}}$. Find the relative maximum and relative minimum by applying the second derivative test.

[5 Marks]

Part B
(75 marks)

1. a) Given that $z = 1 - i$, express $\frac{z + 2 - i}{z^2 - i}$ in the form $a + bi$ where a and b are real numbers. [5 marks]

- b) Given that the complex number z and its conjugate $\bar{z}$ satisfy the equation $\frac{\bar{z}}{z} + \frac{1}{\bar{z}} = \frac{2 - i}{i}$, find the possible values of z . [7 marks]

2. Solve the equations
 - a) $5^{2x} + 4(5^x) - 5 = 0$ [5 marks]

 - b) Find the solution set for $\left| \frac{2x}{x+3} \right| < 3$ [7 marks]

3. Given $f(x) = 3x^2 + 2x$ and $g(x) = x^2 - 2$.
 - a) Find $(f \circ g)(x)$. Hence, factorise $(f \circ g)(x)$ completely. [3 marks]

 - b) The function h is defined by $h(x) = \frac{g(x)}{(f \circ g)(x)}$. Find the values of x such that $h(x)$ is undefined. [5 marks]

 - c) Given $(k \circ g \circ h)(x) = \left(\frac{1}{3x^2 - 4} \right)^2 + 1$, find the function of $k(x)$. [5 marks]

4. a) A function f is defined as $f(x) = \begin{cases} 2|x+1| & x < 0 \\ 4 & x = 0 \\ 2e^{-2x} & x > 0 \end{cases}$.
Find $\lim_{x \rightarrow 0} f(x)$. Hence, determine whether $f(x)$ is continuous at $x = 0$. [5 Marks]

- b) Determine the vertical asymptote of $y = \frac{2x-7}{x^2-4x}$ [3 Marks]

5. a) If $y = e^{-2x} \ln x$, show that $x^2 \left[\frac{d^2y}{dx^2} + 2 \frac{dy}{dx} \right] + e^{-2x}(2x + 1) = 0$ [7 Marks]

b) Given that $x = 2t - \frac{1}{t}$ and $y = 2t + \frac{1}{t}$, find the value of $\frac{dy}{dx}$ at $t = -3$ [4 Marks]

6. An open can has a volume of 20cm^3 . Find its dimension (r, h), so that its surface area is minimum [7 Marks]

7. A water through of length 1 m has a triangular cross-section in the shape of an equilateral triangle of side 60 cm. Water is filled to the through at a rate of $50\sqrt{3} \text{ cms}^{-1}$. Find the rate of increase of the depth of the water in the through when it has been filled for 10 minutes.

[12 marks]

End of questions

Answers.

Part A

1. a. 10 , b. 4 c. $\frac{\sqrt{2}}{2}$
2. i. $\frac{dy}{dx} = 6xe^{3x^2-2}$ ii. $\frac{dy}{dx} = \frac{2(\sin 2x + \cos 2x)}{\sin 2x - \cos 2x}$ iii. $\frac{dy}{dx} = \frac{-(ax+hy)}{(hx+by)}$
3. (- 4 , 15.12) is a maximum point , (0 , 0) is a minimum point

Part B

1. a) $\frac{2}{3} + i$
b) $z = -\frac{1}{2}i$ and $z = -\frac{1}{2} - \frac{1}{2}i$
2. a) $x = 0$
b) $x = 3, x = \frac{1}{2187}$
3. a) $(f \circ g)(x) = (x - \sqrt{2})(x + \sqrt{2})(\sqrt{3}x - 2)(\sqrt{3}x + 2)$
b) $x = -\frac{2}{\sqrt{3}}$ and $x = \frac{2}{\sqrt{3}}$
c) $k(x) = x + 3$
4. a. 2 , Not continuous
b. $x = 0$, $x = 4$
5. b) $\frac{17}{19}$
6. Surface area is minimum when $r = 1.85$ cm and $h = 1.86$ cm
7. $\frac{1}{40} \text{ cms}^{-1}$